

The Logarithmic Upper Bound Problem

Precise formulation and a sufficient tail criterion

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Fix

$$P = (\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8}), \quad Q = (\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64}).$$

An exact depth- n zero-repair tree consists of probability laws B_h on $\mathbb{Z}_{\geq 0}$ with finite first moments, satisfying

$$Q * B_h = \sum_{x=0}^4 P(x) \delta_x * B_{hx} \quad (|h| < n). \quad (1)$$

Write $\mathcal{B}_n$ for the minimum root mean and put $\mathcal{B}_0 = 0$. All logarithms below are natural.

Open problem 1 (Main asymptotic question). *Determine whether*

$$\sup_{n \geq 2} \frac{\mathcal{B}_n}{\log n} < \infty. \quad (2)$$

Equivalently, does there exist a constant $C < \infty$, independent of n , such that

$$\mathcal{B}_n \leq C \log(n+1) \quad \text{for every } n \geq 1?$$

The established logarithmic lower bound implies that a positive answer would give $\mathcal{B}_n = \Theta(\log n)$. A negative answer would mean precisely

$$\limsup_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} = \infty,$$

or equivalently $\mathcal{B}_n \notin O(\log n)$. This does not by itself assert $\mathcal{B}_n / \log n \rightarrow \infty$.

What is already established

The preceding note, *The Contact Breaks at Horizon Five*, proves

$$\liminf_{n \rightarrow \infty} \frac{\mathcal{B}_n}{\log n} \geq \kappa_0 := \frac{\log(594184/543559)}{\log 16 \log(524288/74273)} \approx 0.0164347253. \quad (3)$$

Pruning and convolution concatenation give

$$\mathcal{B}_n \leq \mathcal{B}_{n+1}, \quad \mathcal{B}_{m+n} \leq \mathcal{B}_m + \mathcal{B}_n.$$

The certified depth-five feasible tree gives $\mathcal{B}_n \leq 0.3783157 \lfloor n/5 \rfloor + \mathcal{B}_{n \bmod 5}$, hence $O(n)$. These facts leave (2) unresolved. In particular, even a proof that $\lim_n \mathcal{B}_n / n = 0$ would not establish logarithmic growth. The contact failure at depth five neither proves nor disproves (2).

A sufficient estimate at successive scales

One possible route is to prove a constant $D < \infty$ such that

$$\mathcal{B}_{2n} \leq \mathcal{B}_n + D \quad (n \geq 1). \quad (4)$$

Indeed, monotonicity and iteration would yield

$$\mathcal{B}_n \leq \mathcal{B}_{2^{\lceil \log_2 n \rceil}} \leq \mathcal{B}_1 + D \lceil \log_2 n \rceil = O(\log n).$$

Estimate (4) is a sufficient condition, not an established bound or a claimed equivalent of the open problem. Ordinary concatenation only gives $\mathcal{B}_{2n} \leq 2\mathcal{B}_n$.

A sufficient condition using an infinite tree

For the following proposition only, allow probability laws in an infinite tree to have infinite first moments. The finite trees produced from it will have bounded support and finite means.

Proposition 1 (Truncated-reserve criterion). *Suppose there is a family $(\hat{B}_h)_{h \in \{0,1,2,3,4\}^{<\infty}}$ of probability laws on $\mathbb{Z}_{\geq 0}$ satisfying (1) at every node. Let $J_0 \sim \hat{B}_\emptyset$. Then, for every $n \geq 1$,*

$$\mathcal{B}_n \leq \mathbb{E}[\min(J_0, 4n)] = \sum_{j=1}^{4n} \mathbb{P}(J_0 \geq j). \quad (5)$$

Consequently, if some constant $A < \infty$ satisfies

$$\mathbb{P}(J_0 \geq j) \leq \frac{A}{j} \quad (j \geq 1), \quad (6)$$

then

$$\mathcal{B}_n \leq AH_{4n} \leq A(1 + \log(4n)), \quad (7)$$

where $H_m = \sum_{j=1}^m 1/j$. Such a construction would answer the main open problem positively.

Proof. The tree equations give a causal realization. Given output history h and reserve $J_t = j$, draw a fresh $Y_{t+1} \sim Q$, independent of the entire past. With $s = j + Y_{t+1}$, draw X_{t+1} using

$$K_h(x | s) = \frac{P(x)\hat{B}_{hx}(s-x)}{(Q * \hat{B}_h)(s)}$$

when the denominator is positive; values on zero-probability events are irrelevant. Set $J_{t+1} = s - X_{t+1}$. Equation (1) normalizes this kernel. Induction gives iid outputs with law P and conditional reserve laws $\hat{B}_h$. No moment assumption is used here.

Fix n and subtract the same random quantity

$$D = (J_0 - 4n)_+ \quad \text{from every } J_t, \quad 0 \leq t \leq n.$$

If $J_0 \leq 4n$, the trajectory is unchanged. Otherwise its shifted reserve is

$$J'_t = 4n + \sum_{i=1}^t (Y_i - X_i) \geq 4n - 4t \geq 0.$$

In both cases $0 \leq J'_t \leq 4n + 4t$. The X, Y trajectory is unchanged, and each fresh Y_{t+1} remains independent of the shifted past, since D depends only on J_0 . Thus the conditional laws of J'_t given the output histories form an exact depth- n tree with bounded support. Its root is $J'_0 = \min(J_0, 4n)$, proving the inequality in (5). The equality is the tail-sum formula for a bounded nonnegative integer random variable. Summing (6) and using $H_m \leq 1 + \log m$ proves (7). $\square$

Proposition 2 (Necessary tail size for this route). *For any infinite family as above, its root tail must satisfy*

$$\limsup_{j \rightarrow \infty} j\mathbb{P}(J_0 \geq j) \geq \kappa_0. \quad (8)$$

In particular, a tail $o(1/j)$ is impossible, and the constant in (6), if such a construction exists, must be at least κ_0 .

Proof. If the limsup in (8) were less than κ_0 , choose $a < \kappa_0$ with $\mathbb{P}(J_0 \geq j) \leq a/j$ for all sufficiently large j . Then (5) would give $\mathcal{B}_n \leq a \log n + O(1)$, contradicting (3). $\square$

Status. The two propositions are implications with proofs. An infinite exact family satisfying (6) has not been constructed here, and (4) has not been proved. The upper bound $O(\log n)$ therefore remains unresolved in this work.